BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) — (GROUP-I) — HSSC-(P-II)-A-2024 (For All Sessions)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8195

Note: Note: Write Answers to the Questions on the objective answer sheet provided. Four possible answers A, B, C and D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with Marker or Pen ink on the answer sheet provided.

Q1. Midpoint of A(2,0), B(0,2) is:

(A) (0,2)
(B) (2,0)
(C) (2,2)
(D) (1,1)

Q2. The ___ point satisfies $x + 2y < 6$

(A) (4,1)
(B) (3,1)
(C) (1,3)
(D) (1,4)

Q3. In a conic, the ratio of the distance from a fixed point to the distance from a fixed line is: [sic]

(A) Focus
(B) Vertex
(C) Ecentricity [sic]
(D) Centre

Q4. Standard equation of Parabola is:

(A) $y^2 = 4ax$
(B) $x^2 + y^2 = a^2$
(C) $x^2/a^2 - y^2/b^2 = 1$
(D) $x^2/a^2 + y^2/b^2 = 1$

Q5. Equation of tangent to circle $x^2 + y^2 = a^2$ at $P(x_1,y_1)$ is:

(A) $xx_1 + yy_1 = a^2$ (B) $xx_1 - yy_1 = a^2$

(C) $xy_1 + yx_1 = a^2$ (D) $xy_1 - yx_1 = a^2$

Q6. The volume of parallelopiped = _____.

(A) $(\underline{u} \times \underline{v}).\underline{\omega}$ (B) $(\underline{u} \times \underline{v}) \times \underline{\omega}$

(C) $\underline{u} \times (\underline{v} \times \underline{\omega})$ (D) $\underline{u} \times (\underline{u} \times \underline{v})$

Q7. The non-zero vectors are perpendicular when:

(A) $\underline{u}.\underline{v} = 1$ (B) $|\underline{u}.\underline{v}| = 1$

(C) $\underline{u}.\underline{v} = 0$ (D) $\underline{u}.\underline{v} \neq 0$

Q8. $\underline{j} \times \underline{k} =$ _____.

(A) $\underline{i}$ (B) $-\underline{i}$

(C) 0 (D) $\underline{k}$

Q9. The range of $f(x) = 2 + \sqrt{(x - 1)}$ is:

(A) $[1,+\infty)$ (B) $[2,+\infty)$

(C) $(1,+\infty)$ (D) $(2,+\infty)$

Q10. The perimeter P of square as a function of its area A:

(A) $3\sqrt{A}$ (B) $4\sqrt{A}$

(C) $\sqrt{A}$ (D) $2\sqrt{A}$

Q11. If $f(x) = 1/x^2$ then $\dot{f}(3) =$ _____.

(A) 1/9 (B) −2/3

(C) −2/27 (D) 1/27

Q12. If $\dot{f}(c) = 0$ & $f''(c) > 0$ then C is point of:

(A) Maxima (B) Minima

(C) Inflection (D) Constant

Q13. $d/dx(\log_a x) =$ _____. [option (C) partially obscured by watermark; as shown]

(A) 1/(xlna) (B) lna/x

(C) 1/x (D) −1/(xlna)

Q14. d/dx(cot ax) = _____.

(A) $\text{cosec}^2 ax$ (B) $a\ \text{cosec}^2 ax$

(C) $-a\ \text{cosec}^2 ax$ (D) $-a\ \text{cosec}\ ax$

Q15. $\int 1/\sqrt{(1-x^2)}\, dx =$ _____.

(A) $\text{Sin}^{-1}x + c$ (B) $\text{Cos}^{-1}x + c$

(C) $-\text{Sin}^{-1}x + c$ (D) $-\text{Cos}^{-1}x + c$

Q16. $\int (1/x)\, dx =$ _____.

(A) $\ln x + c$ (B) $1/x^2 + c$

(C) $-1/x^2 + c$ (D) $1/x + c$

Q17. The solution of differential equation $dy/dx = -y$ is:

(A) $y = xe^{-x}$ (B) $y = ce^{-x}$

(C) $y = e^x$ (D) $y = ce^x$

Q18. $\int_0^1 1/(1+x^2)\, dx =$ _____.

(A) $\pi/4$ (B) $2\pi/3$

(C) $3\pi/4$ (D) π

Q19. x − intercept of the line $2x + 5y - 1 = 0$ is:

(A) 2 (B) 3

(C) 1/2 (D) 1/5

Q20. Slope of y − axis is:

(A) 0 (B) 1

(C) −1 (D) Undefined

Mathematics (Subjective) — (GROUP-I) — HSSC-(P-II)-A/2024 (For All Sessions)

Time Allowed: 2:30 hours | **Maximum Marks:** 80

Q2. Write short answers of any eight parts from the following: (8x2=16)

i If f(x) = 2x + 1, then find fof(x).

ii Express the area A of a circle as a function of its circumference C.

iii Evaluate lim_{h→0} (√(x+h) − √x)/h

iv Define continuous function.

v Differentiate $(\sqrt{x} + 1/\sqrt{x})^2$ w. r. t x

vi Find dy/dx if $y^2 - xy - x^2 + 4 = 0$

vii Differentiate $x^2\sec 4x$ w. r. t x

viii Differentiate $\sin^2 x$ w. r. t. $\cos^4 x$

ix Find $\dot{f}(x)$ if $f(x) = e^x(1 + \ln x)$

x Find y_2 if $y = \ln(x^2 - 9)$

xi Prove that $\ln(1 + x) = x - x^2/2 + x^3/3 - x^4/4 + \cdots$

xii Determine the interval in which $f(x) = \cos x$ is decreasing; $x \in (-\pi/2, \pi/2)$.

Q3. Write short answers of any eight parts from the following: (8x2=16)

i Solve the differential equation $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$

ii Find the area between x − axis and the curve $y = x^2 + 1$ from $x = 1$ to $x = 2$

iii Evaluate: $\int_1^e x \ln x\, dx$

iv Evaluate the integral $\int (-2x)/\sqrt{(4-x^2)}\, dx$

v Evaluate: $\int(\sqrt{x} - 1/\sqrt{x})^2\, dx$

vi Evaluate the integral ∫(a + 2x)^{3/2} dx [as printed: (a ⋈ 2x)^{3/2}]

vii Find the approximate change in the volume of a cube if length of its each edge changes from 5 to 5.02.

viii Show that the points A(0,2), B(√3,−1) and C(0,−2) are vertices of a right triangle.

ix Convert the equation of line $4x + 7y + 2 = 0$ into normal form.

x Fine the angle from the line with slope −7/3 to the line with slope 5/2. [sic]

xi Find the pair of lines represented by $3x^2 + 7xy + 2y^2 = 0$.

xii Find the point of intersection of lines $3x + y + 12 = 0$ and $x + 2y - 1 = 0$.

Q4. Write short answers of any nine parts from the following: (9x2=18)

i Define feasible region.

ii Graph the solution set of in-equality $3x + 7y \geq 21$. [sic]

iii Find equation of circle with ends of diameter at (−3,2) and (5,−6).

iv Write down equation of tangent to the circle $x^2 + y^2 = 25$ at $(5\cos\theta, 5\sin\theta)$

v Find focus and vertex of Parabola $x^2 = 4(y - 1)$

vi Find equation of ellipse with data Foci (±3, 0) Minor axis of length 10.

vii Find center of hyperbola $x^2 - y^2 + 8x - 2y - 10 = 0$

viii Find equation of Normal to $y^2 = 4ax$ at $(at^2, 2at)$.

ix Find the sum of vector $\vec{AB}$ and $\vec{CD}$ given four points A(1,−1) B(2,0) C(−1,3) and D(−2,2)

x Find ∝, so that $|\propto \underline{i} + (\propto + 1)\underline{j} + 2\underline{k}| = 3$

xii If $\underline{v}$ is a vector for which $\underline{v}.\underline{i} = 0$ $\underline{v}.\underline{j} = 0$ $\underline{v}.\underline{k} = 0$, find $\underline{v}$ [as labeled in print]

xii Find the area of triangle determined by the points P(0,0,0) Q(2,3,2) and R(−1,1,4)

xiii Find the value of $2\underline{i} \times 2\underline{j}.\hat{k}$

Q. SECTION-II

Note Attempt any three questions. Each question carries equal marks: (10x3=30)

Q5. (a) Find the values of m and n, so that given function f is continuous at x = 3 when f(x) = { mx, if x < 3; n, if x = 3; −2x + 9, if x > 3 } (05)

(b) Find dy/dx, when $x = a(1-t^2)/(1+t^2)$, $y = 2bt/(1+t^2)$ (05)

Q6. (a) If $y = (\cos^{-1}x)^2$, prove that $(1 - x^2)y_2 - xy_1 - 2 = 0$. (05)

(b) Evaluate the integral $\int e^x \sin x \cos x \, dx$. (05)

Q7. (a) Solve the differential equation $y - x\,dy/dx = 3(1 + x\,dy/dx)$. (05)

(b) Graph the feasible region and corner points of the inequalities $2x + y \leq 10$; $x + 4y \leq 12$; $x + 2y \leq 10$; (05)

Q8. (a) Show that the circles: $x^2 + y^2 + 2x - 8 = 0$; $x^2 + y^2 - 6x + 6y - 46 = 0$ touch internally. (05)

(b) Using vector method, for any triangle ABC, prove that: $c^2 = a^2 + b^2 - 2ab\cos C$. (05)

Q9. (a) Find the focus, vertex and directrix of the Parabola; $x^2 = 4(y - 1)$ (05)

(b) Find the lines represented by $3x^2 + 7xy + 2y^2 = 0$ and also find measure of the angle between them. (05)

GROUP – II

Mathematics (Objective) — (GROUP-II) — HSSC-(P-II)-A-2024 (For All Sessions)

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 8194

> Note: Note: Write Answers to the Questions on the objective answer sheet provided. Four possible answers A, B, C and D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with Marker or Pen ink on the answer sheet provided.

Q1. If r = 0, the circle is called:

(A) Unit circle
(B) Circle
(C) Ellipse
(D) Point circle

Q2. $[\underline{i}\ \underline{j}\ \underline{k}]$ =

(A) $\underline{i}$
(B) $-\underline{i}$
(C) 1
(D) 0

Q3. If $\underline{u} = 2\underline{i} - \underline{j} + \underline{k}$, $\underline{v} = 4\underline{i} + 2\underline{j} - \underline{k}$ then $\underline{u} \times \underline{u}$ = [sic]

(A) u^2
(B) 0
(C) 1
(D) 2

Q4. If $\underline{u}$, $\underline{v}$ are two non-zero vectors, then area of parallelogram =

(A) $|\underline{u} \times \underline{v}|$
(B) $1/2|\underline{u} \times \underline{v}|$
(C) $1/6|\underline{u} \times \underline{v}|$
(D) $1/2(\underline{u} \times \underline{v})$

Q5. If k is any real number, lim_{x→a}[kf(x)] =

(A) lim_{x→a} f(x)
(B) lim_{x→a} k
(C) k lim_{x→a} f(x)
(D) f(x)

Q6. If f(x) = x + 3 then: lim_{x→3} f(x) =

(A) 6
(B) 0
(C) −3
(D) 3

Q7. If y = e^{f(x)} then dy/dx = [sic: options (B) and (C) duplicate]

(A) e^{f(x)}
(B) f(x)e^{f(x)}
(C) f(x)e^{f(x)}
(D) $\dot{f}$(x)e^{f(x)}

Q8. Derivative of $x\sqrt{(x^2 + 3)}$ w. r. t x is:

(A) $(2x^2 + 3)/\sqrt{(x^2 + 3)}$
(B) $3x/(2\sqrt{(x^2 + 3)})$
(C) $(3x^2 + 3)/(x\sqrt{(x^2 + 3)})$
(D) $(3x^2 + 3)/(2x\sqrt{(x^2 + 3)})$

Q9. Derivative of $\tanh(x^2)$ is:

(A) $2x\ \text{sech}^2 x$
(B) $2\ \text{sech}^2 x^2$
(C) $2x\ \text{sech}^2 x^2$
(D) $\text{sech}^2 x^2$

Q10. Derivative of "x" w. r. t "x" is: [sic]

(A) x^2
(B) 2
(C) 0
(D) 1

Q11. In integration, substitution of $\sqrt{(4 - x^2)}$ is:

(A) $x = \sin\theta$ (B) $x = 2\sin\theta$

(C) $x = \sin 2\theta$ (D) $x = 2\cos\theta$

Q12. $\int \text{Tan } x\, dx$ = [sic: option (B) partially struck through as shown]

(A) ln cos x + c (B) 1/ln cosx ⋯ c [as shown]

(C) $-\ln|\cos x| + c$ (D) $\text{Sec}^2 x + c$

Q13. Solution of differential equation: $(e^x + e^{-x})\, dy/dx = e^x - e^{-x}$ is:

(A) $-\ln(e^x + e^{-x}) + c$ (B) $\ln(e^x - e^{-x}) + c$

(C) $\ln(e^x + e^{-x}) + c$ (D) $(e^x + e^{-x})^2/2$

Q14. $\int \text{Sinx Cos } x\, dx$ =

(A) $\text{Sin}^2 x/2 + c$ (B) $\text{Cos}^2 x/2 + c$

(C) $-\text{Sin } x + c$ (D) $\text{Cosx} + c$

Q15. The line: $ay + b = 0$ is

(A) Parallel to y-axis (B) Parallel to x-axis

(C) Passing through origin (D) Lies in Quad. I

Q16. The slope of line joining the points $(-2,4)$; $(5,11)$ is:

(A) 1 (B) -1

(C) $45°$ (D) $-45°$

Q17. The location of the plane of the point P(x,y) for which $y = 0$ at:

(A) Origin (B) y − axis

(C) x − axis (D) Ist Quad

Q18. The maximum and minimum values occur at:

(A) Corner point (B) Any point

(C) Convex region (D) Corner points of feasible region

Q19. The line intersect the circle at:

(A) One point (B) Two points

(C) Infinite points (D) More than two points

Q20. Diameter of circle: $x^2 + y^2 = 16$ is:

(A) 8 (B) 4

(C) 16 (D) 32

Mathematics (Subjective) — (GROUP-II) — HSSC-(P-II)-A/2024 (For All Sessions)

Time Allowed: 2:30 hours | **Maximum Marks:** 80

Q2. Write short answers of any eight parts from the following: (8x2=16)

i Define even function with example.

ii Find fog(x) if f(x) = 2x + 1, g(x) = 3/(x−1), x ≠ 1.

iii Evaluate: lim_{x→2} (√x − √2)/(x−2)

iv Prove that Sinh2x = 2 Sinhx Coshx.

v Find dy/dx from first principles if y = 1/√(x+a).

vi Differentiate w. r. t x; $(x^2+1)^2/(x^2-1)$.

vii Find dy/dx if $x^2 - 4xy - 5y = 0$.

viii Differentiate w. r. t θ; $\tan^3\theta \sec^2\theta$.

ix Find $\dot{f}(x)$ if $f(x) = x^3 e^{1/x}$.

x Find y_2 if $y = 2x^5 - 3x^4 + 4x^3 + x - 2$.

xi Apply Maclaurin Series expansion to prove that: $\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \cdots$

xii Find extreme values for $f(x) = 3x^2$.

Q3. Write short answers of any eight parts from the following: (8x2=16)

i Evaluate $\int x\sqrt{(x^2 - 1)}\, dx$

ii Use differentials to approximate the value of $(31)^{1/5}$

iii Evaluate: $\int x/\sqrt{(4+x^2)}\, dx$

iv Evaluate the integral $\int e^{m \tan^{-1}x}/(1+x^2)\, dx$

v Evaluate: $\int_1^2 x/(x^2 + 2)\, dx$

vi Find the area between x − axis and the curve $y = 4x - x^2$

vii Solve the differential equation $1/x\, dy/dx = 1/2(1 + y^2)$

viii The points A(−5,−2) and B(5,−4) are ends of a diameter of a circle. Find the centre and radius of circle.

ix The coordinates of a point p are (−6,9). The axes are translated through the point O(−3,2). Find the coordinates of p referred to the new axes

x Check whether the origin and the point p(5,−8) lies on the same side or on the opposite sides of the line 3x + 7y + 15 = 0

xi By means of slopes, show that the following points lie on the same line (−4,6); (3,8); (10,10).

xii Determine the value of p such that the lines $2x - 3y - 1 = 0$, $3x - y - 5 = 0$ and $3x + py + 8 = 0$ meet at a point.

Q4. Write short answers of any nine parts from the following: (9x2=18)

i Graph the solution set of $3y - 4 \leq 0$ in xy − plane.

ii Define convex region.

iii Find an equation of circle of radius a and lying in the second quadrant tangent to both the axes.

iv Find center and radius of circle $5x^2 + 5y^2 + 24x + 36y + 10 = 0$.

v Write down equation of normal to the circle $x^2 + y^2 = 25$ at (4,3).

vi Find vertex and directrix of the parabola $y^2 = -12x$.

vii Find the point of intersection of conics $x^2 + y^2 = 8$ and $x^2 - y^2 = 1$.

viii Find center and foci of hyperbola $y^2/4 - x^2 = 1$.

ix Find a vector of magnitude 4 and is parallel to $2\underline{i} - 3\underline{j} + 6\underline{k}$.

x Find direction cosines of $\overrightarrow{PQ}$ where P = (2,1,5) and Q = (1,3,1).

xi Find volume of parallelopiped whose edges are $\underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}$, $\underline{v} = 2\underline{i} - \underline{j} - \underline{k}$ and $\underline{w} = \underline{i} + \underline{k}$.

xii Find the value of $[\underline{k}\ \underline{i}\ \underline{j}]$.

xiii Find $\propto$ so that $\underline{u} = \propto\underline{i} + 2\propto\underline{j} - \underline{k}$ and $\underline{v} = \underline{i} + \propto\underline{j} + 3\underline{k}$ are perpendicular.

Q. SECTION-II

Note Attempt any three questions. Each question carries equal marks: (10x3=30)

Q5. (a) Evaluate: $\lim_{\theta\to 0} (\tan\theta - \sin\theta)/\sin^3\theta$ (b) Differentiate $\cos\sqrt{x}$ from the first principle. (5+5)

Q6. (a) Show that $y = \ln x/x$ has maximum value at x = e (b) Evaluate: $\int x^3 \cos x\, dx$ (5+5)

Q7. (a) Evaluate: $\int_{\pi/6}^{\pi/2} \cos x/(\sin x(2 + \sin x))\, dx$ (b) Minimize $z = 2x + y$ subject to constraints $x + y \geq 3$; $7x + 5y \leq 35$; $x \geq 0$; $y \geq 0$ (5+5)

Q8. (a) Find the coordinates of the points of intersection of the line $x + 2y = 6$ with the circle: $x^2 + y^2 - 2x - 2y - 39 = 0$ (5)

(b) If $\underline{a} = 4\underline{i} + 3\underline{j} + \underline{k}$ and $\underline{b} = 2\underline{i} - \underline{j} + 2\underline{k}$. Find a unit vector perpendicular to both $\underline{a}$ and $\underline{b}$. Also find the sine of the angle between them. (5)

Q9. (a) Find the focus, vertex and directrix of the Parabola $x + 8 - y^2 + 2y = 0$ (5)

(b) Find coordinates of the circumcenter of the triangle whose vertices are $A(-2,3)$, $B(-4,1)$ and $C(3,5)$. (5)